Student:	Email:
jeffar alaboudi	jmalab01@louisville.edu

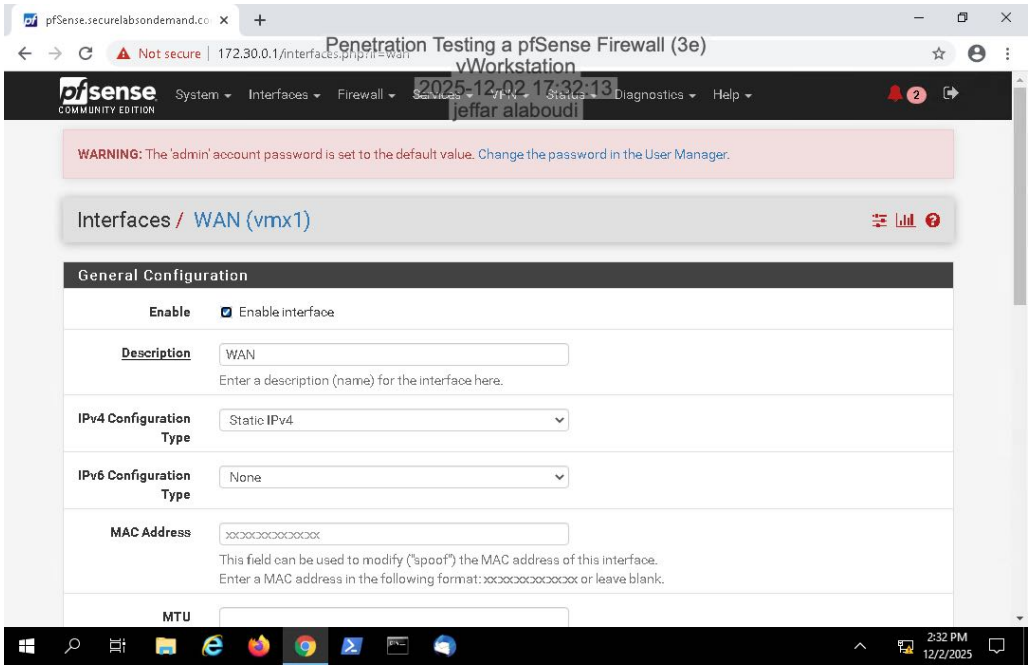
Time on Task:	Progress:
0 hours, 13 minutes	100%

Report Generated: Tuesday, December 2, 2025 at 5:39 PM

Section 1: Hands-On Demonstration

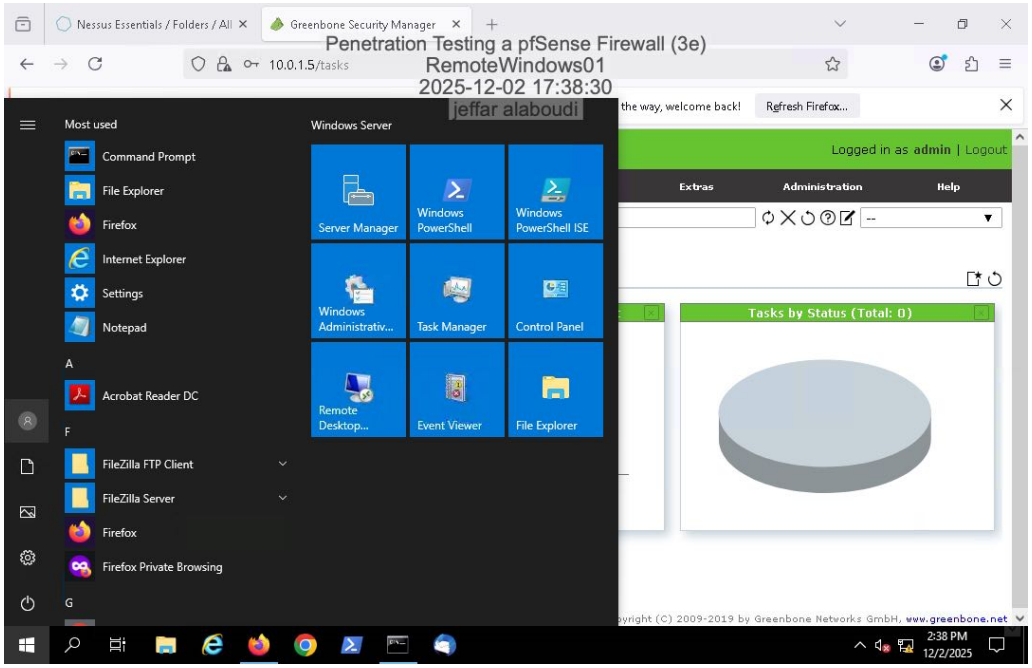
Part 1: Examine a pfSense Firewall Configuration

12. **Make a screen capture** showing the **WAN rules table**.

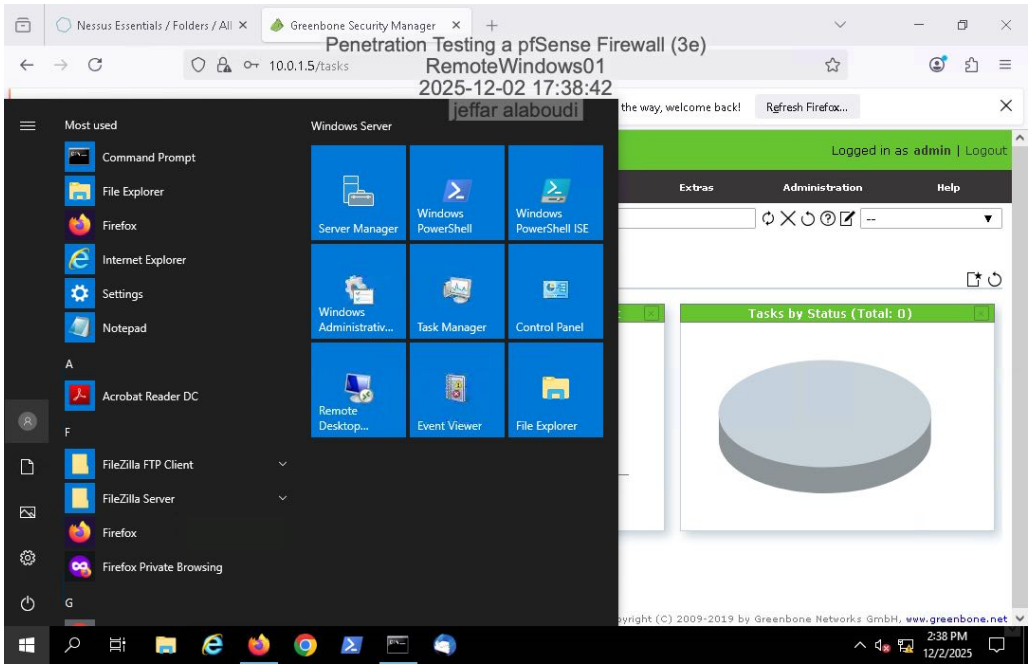


Part 2: Conduct a Penetration Test on the Network

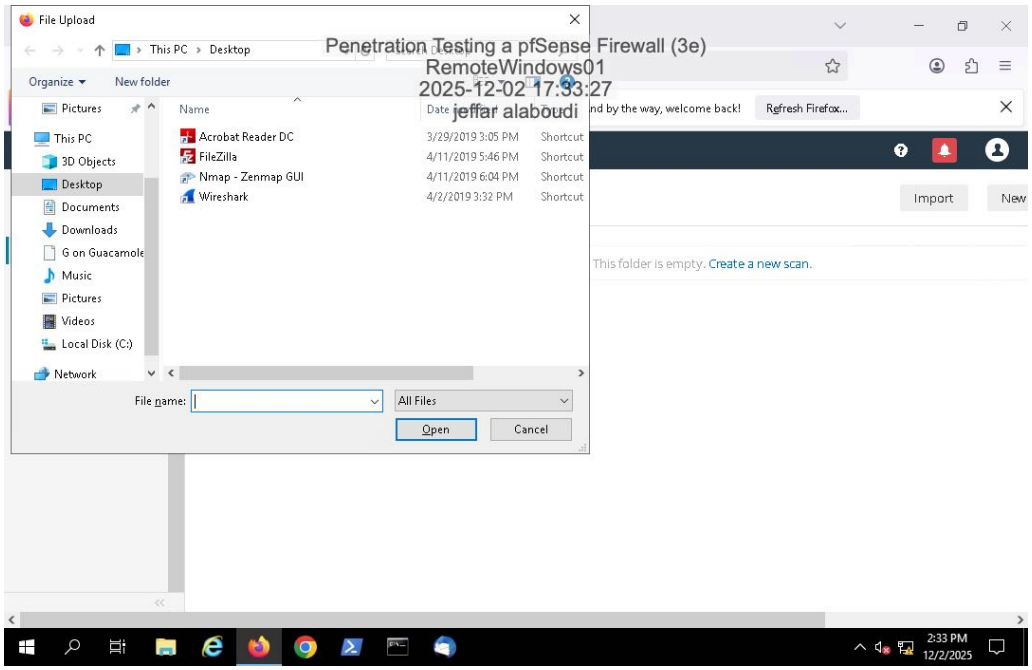
11. **Make a screen capture** showing the ***yourname*** **pen test scan results**.



13. **Make a screen capture** showing the **list of vulnerabilities**.



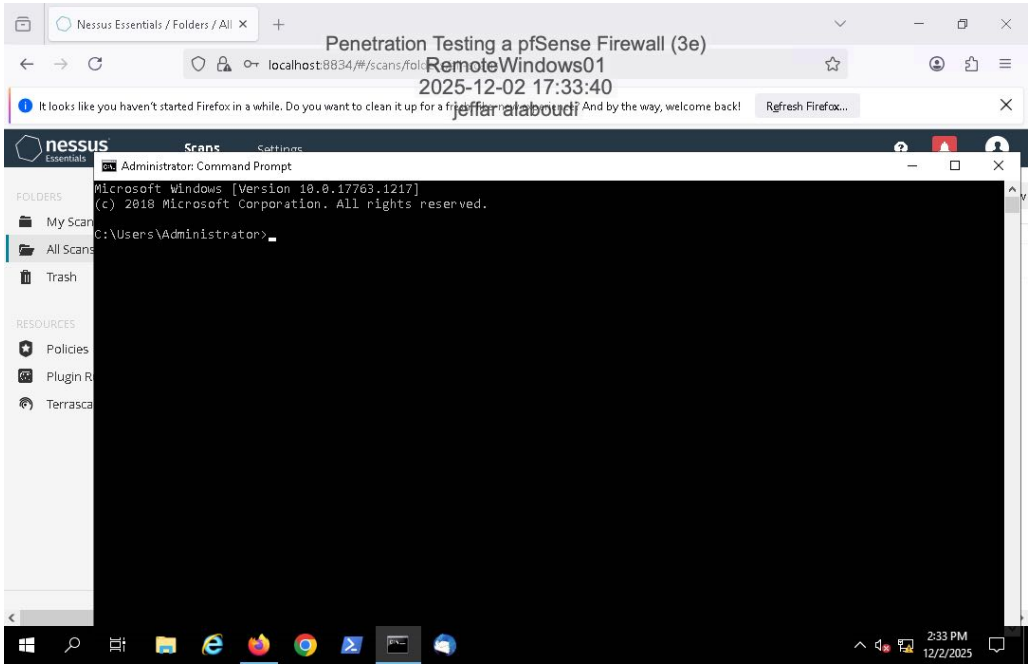
30. **Make a screen capture** showing the **updated vulnerability report summary**.



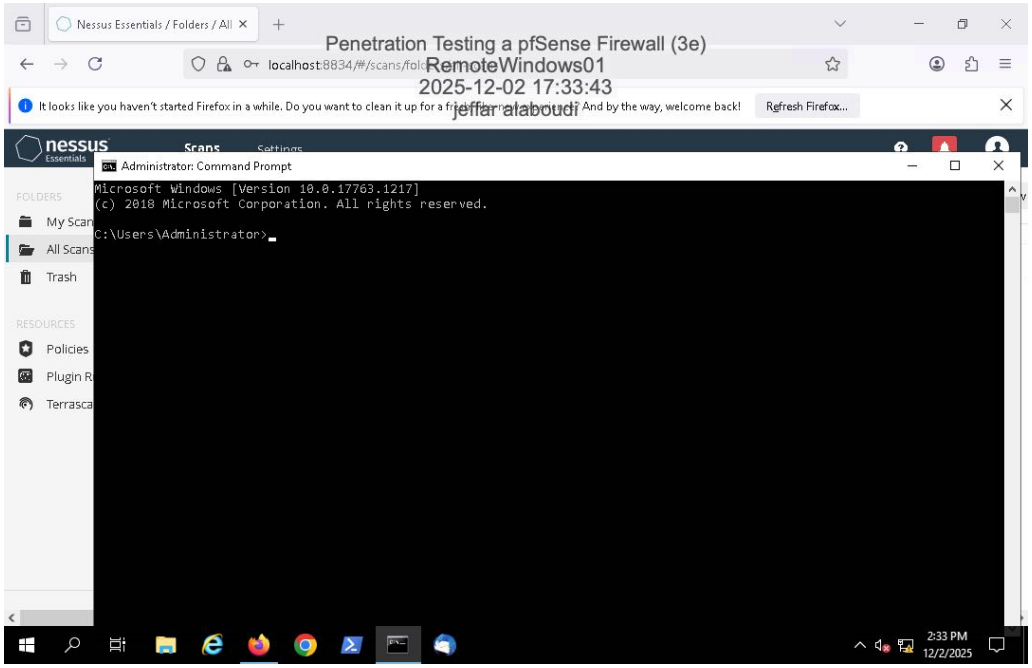
Section 2: Applied Learning

Part 1: Conduct a Port Scan on the Network

7. **Make a screen capture** showing the **results of the traceroute command**.

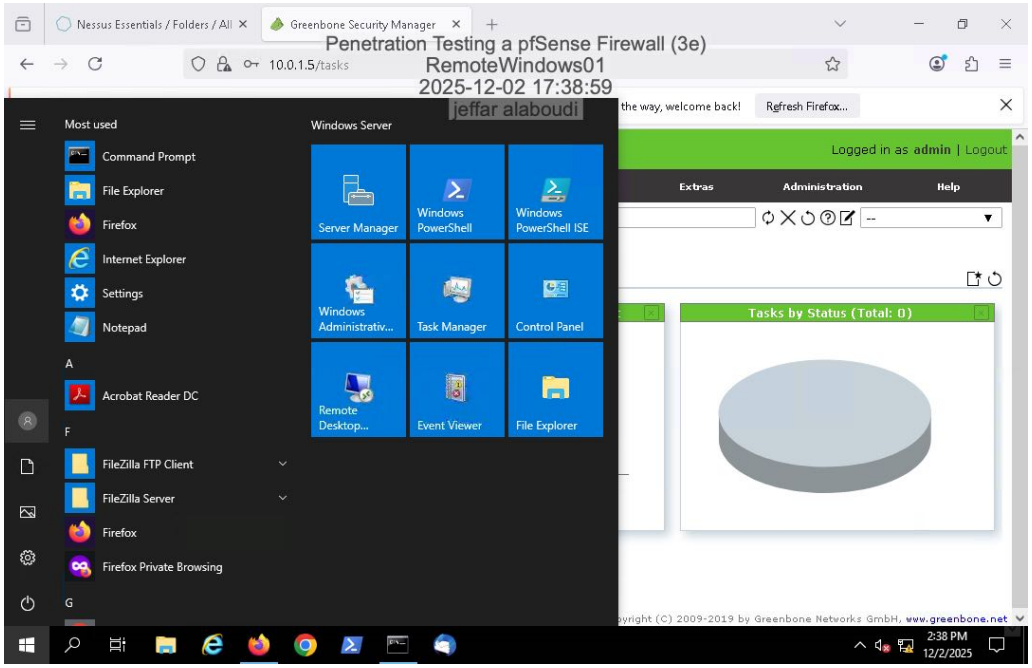


11. **Make a screen capture** showing the result of the **nmap scan with OS detection activated**.

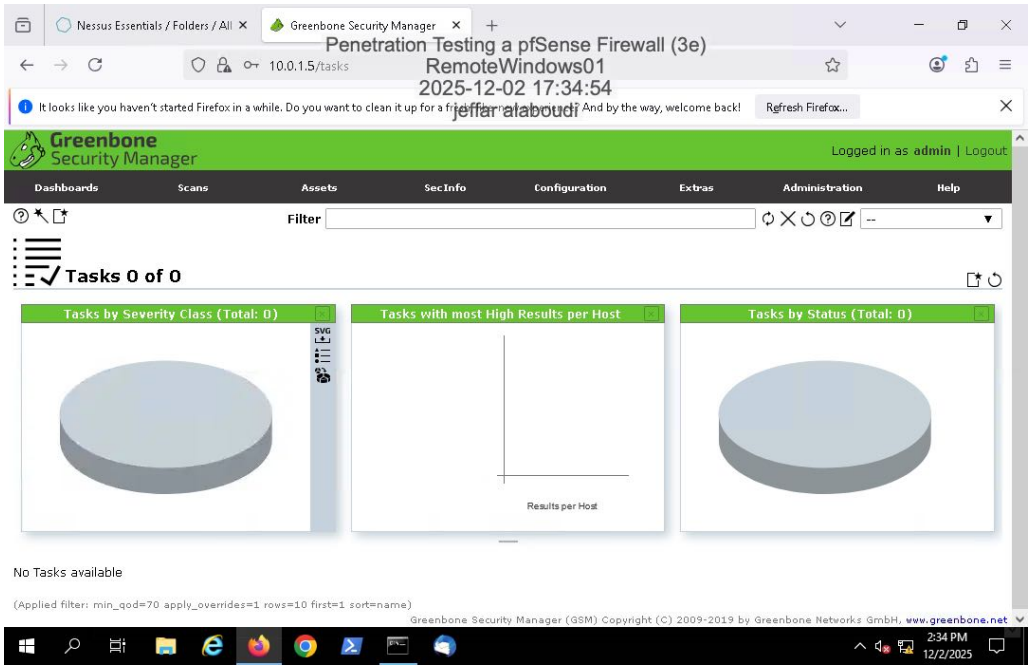


Part 2: Conduct a Vulnerability Scan on the Network

12. **Make a screen capture** showing the **OpenVAS scan report**.



14. **Make a screen capture** showing the **detailed OpenVAS scan results**.



Section 3: Challenge and Analysis

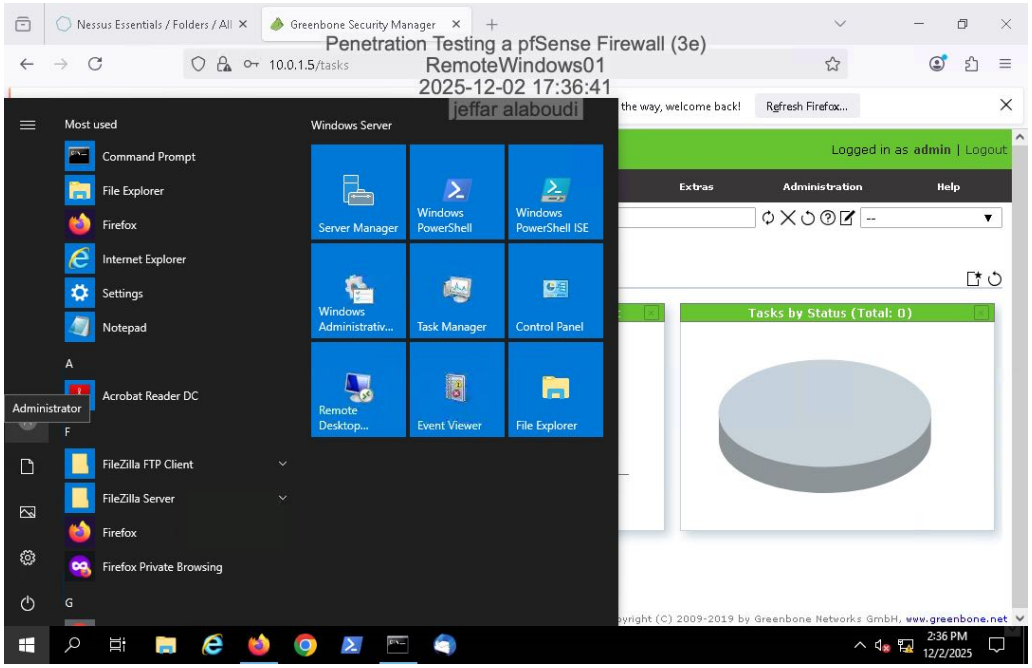
Part 1: Research DMZ Deployment Best Practices

Before beginning the technical portion of your penetration test, you decide to spend some time brushing up on best practices and common mistakes for DMZ deployments - both the network aspect and the servers located therein. Use the Internet to **research** DMZ deployments, then **identify** three best practices and one potential mistake or vulnerability.

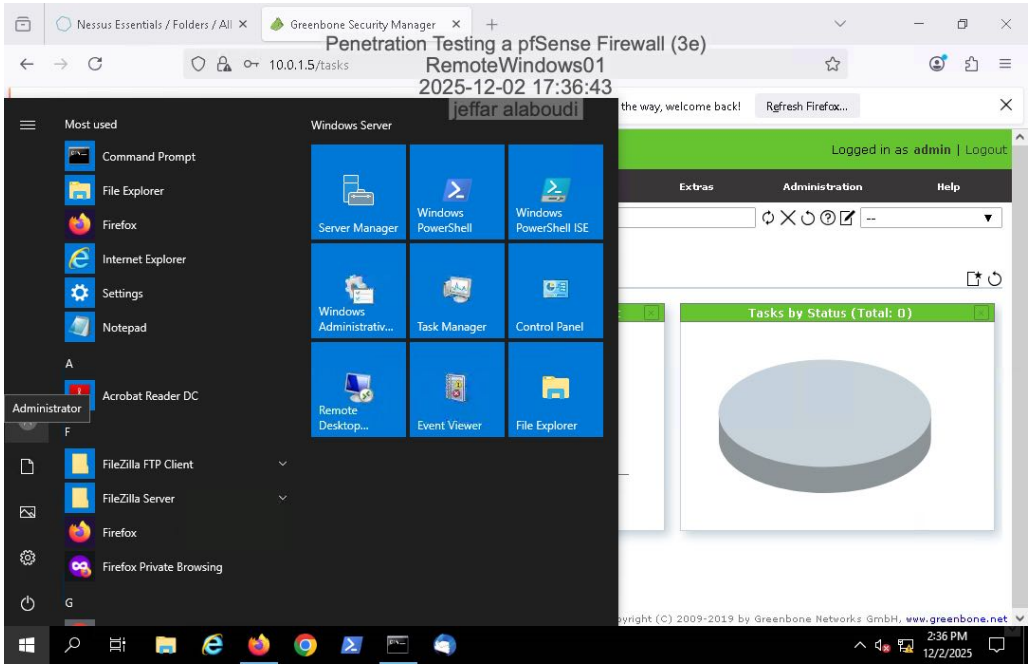
A properly designed DMZ (demilitarized zone) is essential for protecting internal networks from internet-facing threats. Three important best practices include fully isolating the DMZ from the internal LAN so that exposed services cannot directly interact with sensitive internal hosts, using a dual-firewall architecture to separate the internet-to-DMZ path from the DMZ-to-LAN path for layered security, and regularly hardening and updating all DMZ servers since they face a high level of exposure and are frequent targets for exploitation. A common mistake, however, is misconfiguring firewall rules or allowing overly permissive access between the DMZ and the internal network. This single error often eliminates the entire purpose of the DMZ, making it easy for an attacker who compromises a DMZ host to move laterally into the internal network.

Part 2: Conduct a Penetration Test on the DMZ

Make a screen capture showing the **open ports on the corporationtechs.com web server and the DMZ firewall interface**.



Make a screen capture showing the **vulnerability scan results**.



Part 3: Recommend Changes to the DMZ

Based on your research in Part 1 and your findings in Part 2, **prepare a brief summary** of recommended changes that Secure Labs on Demand should make to their DMZ deployment. Remember, your recommendations should apply to both the network configuration and the web server.

Based on the research and the vulnerabilities identified, Secure Labs on Demand should strengthen both their DMZ network design and their web server configuration. At the network level, they should fully isolate the DMZ from the internal LAN by tightening firewall rules and ensuring that only essential, one-way traffic is allowed. Implementing a dual-firewall architecture or stricter segmentation would provide layered protection in case a DMZ system is compromised. They should also review and remove any overly open ports or services that increase the attack surface. On the web server side, the organization should apply hardening steps such as disabling unused services, enforcing strong authentication, and ensuring all software and frameworks are up to date. Logging, monitoring, and regular vulnerability scanning should be enabled to quickly detect abnormal behavior. Together, these improvements would reduce lateral-movement risks, limit exposure of internal resources, and ensure that the DMZ truly functions as a secure buffer between the public internet and the internal network.